A. Painting With Two Colors

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 800 | Tags: constructive algorithms, implementation, math |
https://codeforces.com/problemset/problem/2134/A

You are given three positive integers n, a, and b ($1 \le a,b \le n$).

Consider a row of n cells, initially all white and indexed from 1 to n. You will perform the following two steps **in order**:

1. Choose an integer x such that $1 \le x \le n - a + 1$, and paint the a consecutive cells x, $x+1$, ..., $x+a-1$ red.
2. Choose an integer y such that $1 \le y \le n - b + 1$, and paint the b consecutive cells y, $y+1$, ..., $y+b-1$ blue.

If a cell is painted both red and blue, its final color is blue.

A coloring of the grid is considered *symmetric* if, for every integer i from 1 to n (inclusive), the color of cell i is the same as the color of cell $(n+1-i)$. Your task is to determine whether there exist integers x and y such that the final coloring of the grid is symmetric.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 500$). The description of the test cases follows.

The first and the only line of each test case contains three integers n, a, and b ($1 \le n \le 10^9$, $1 \le a,b \le n$) — the number of cells of the grid and the number of cells to be painted in each step.

Output

For each testcase, output "YES" if it is possible that the final coloring of the grid is symmetric; otherwise, output "NO".

You can output the answer in any case (upper or lower). For example, the strings "yEs", "yes", "Yes", and "YES" will be recognized as positive responses.

Example

Input
7 5 3 1 4 1 2 7 7 4 8 3 7 1 1 1 1000000000 1000000000 1000000000 3 2 1

Output

```
YES
YES
NO
NO
YES
YES
NO
```

Note

In the first test case, the grid becomes symmetric when choosing $x = 2$ and $y = 3$, so the answer is "YES". The following figure illustrates each step of the coloring process:



In the second test case, the grid becomes symmetric when choosing $x = 2$ and $y = 2$, so the answer is "YES". The following figure illustrates each step of the coloring process:



In the third and fourth test cases, it can be proved that no choice of x and y results in a symmetric grid, so the answer is "NO".